The Groups of Order Sixteen Made Easy

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1. INTRODUCTION. There are many introductory texts on group theory that, in particular, classify all groups of order at most fifteen. What about order sixteen? Well, there are fourteen of them, and they do of course make their appearance in higher level texts such as [**1**] or [**7**]. However, to the author's best knowledge, the classification of the groups of order sixteen is always obtained as a special case of a sophisticated theory of p-groups, and many details are left to the reader to verify. This is annoying since it, for example, obstructs the teaching of an advanced undergraduate course devoted to the classification of all groups of order no larger than 23 or even 31.

Here we give a complete classification of the groups of order sixteen that is based on elementary facts. More specifically, we look at extensions of order eight groups N by C_2, the cyclic group of order two. A crucial role is played by Lemma 2, which allows us to discard three of the five groups of order eight. We also deal a lot with automorphisms of groups. In this article no Sylow-theory, no theory of p-groups, no relatively free groups (generators and relations), no group actions, not even the structure of finite Abelian groups will be invoked. The most "difficult" prerequisite we take for granted is Fact 5 about conjugacy classes and centralizers. Semidirect products appear as an additional means to describe the fourteen groups of order sixteen. Yet no knowledge of semidirect products is required; they do not appear in the proofs.

2. PRELIMINARIES AND STATEMENT OF THE MAIN THEOREM. We review some basic concepts and facts for later use. The proofs can be found in most introductory books such as [**8**] or lower level texts. A bijective homomorphism $\phi : G \to H$ of groups is an *isomorphism*; if $H = G$ it is an *automorphism*. We write $G \simeq H$ if there is an isomorphism $\phi : G \to H$, write $H \leq G$ if H is a subgroup of G, and use $H \lhd G$ to signify that H is a normal subgroup of G. For any subsets X and Y of G we put $XY = \{xy : x \in X, y \in Y\}$. If G is generated by $x, y, \ldots, z$, we write $G = \langle x, y, \ldots, z\rangle$. In particular, when $G = \langle x\rangle$ then G is *cyclic*. The *order* of G is its cardinality $|G|$, and $o(x)$ is the order of $\langle x\rangle$. We let $C_n = \{e, x, x^2, \ldots, x^{n-1}\}$ be the cyclic group of order n.

Fact 1. If H_1 and H_2 are subgroups of G such that $H_1 \cap H_2 = \{e\}$ and $xy = yx$ for all x in H_1 and y in H_2, then H_1H_2 is a subgroup isomorphic to the direct product $H_1 \times H_2$.

Fact 2. If $o(x) \leq 2$ for each x in G, then G is Abelian. If in addition $|G| < \infty$, then $G \simeq C_2 \times C_2 \times \cdots \times C_2$.

Indeed, for all x and y in G one has $(xy)(xy) = e$ and $(xy)(yx) = xy^2x = x^2 = e$. Cancelling xy on the left gives $xy = yx$. Using Fact 1, commutativity, and induction on $|G|$ it is straightforward to see that $G \simeq C_2 \times \cdots \times C_2$ in the finite case. In particular $|G| = 4$ implies that $G \simeq C_4$ or $G \simeq C_2 \times C_2$. We write K_4 for the *Klein four group* $C_2 \times C_2$.

The set Aut(G) of all automorphisms of a group G is a group under composition of mappings. Each ϕ in Aut(G) is determined by its values on any set of generators of G. If $C_4 = \langle x \rangle$, then $\mathrm{Aut}(C_4) = \{\varphi_1, \varphi_2\}$, and if $C_8 = \langle x \rangle$, then $\mathrm{Aut}(C_8) = \{\phi_1, \phi_2, \phi_3, \phi_4\}$, where the mappings are described in Fact 3:

Fact 3.

$\mathrm{Aut}(C_4)$	effect on x
φ_1	x
φ_2	$x^3 = x^{-1}$

$\mathrm{Aut}(C_8) \simeq K_4$	effect on x
ϕ_1	x
ϕ_2	x^3
ϕ_3	x^5
ϕ_4	$x^7 = x^{-1}$

All nonidentity elements of $\mathrm{Aut}(C_8)$ have order two. For instance,

$$(\phi_3 \circ \phi_3)(x) = \phi_3\big(\phi_3(x)\big) = \phi_3(x^5) = \big(\phi_3(x)\big)^5 = (x^5)^5 = x^{25} = x.$$

For notational convenience we view the direct product of $C_4 = \langle x \rangle$ and $C_2 = \langle y \rangle$ as $\{e, x, x^2, x^3, y, xy, x^2y, x^3y\}$. With this convention $(x^2y)(x^3y) = x^2x^3yy = x$ is computed "component-wise," as if we had ordered pairs (x^2, y) and (x^3, y). Because this group will appear frequently in this article we christen it K_8, which is nicer than $C_4 \times C_2$. Since each ψ in $\mathrm{Aut}(K_8)$ has $\psi(x)$ of order four, $\psi(y)$ of order two, and $\psi(y) \notin \langle \psi(x) \rangle$, there can be at most eight automorphisms; namely, those indicated by Fact 4:

Fact 4.

$\mathrm{Aut}(K_8) \simeq D_8$	effect on x	effect on y	order of automorphism
ψ_1	x	y	1
ψ_2	x^3y	x^2y	4
ψ_3	x^3	y	2
ψ_4	xy	x^2y	4
ψ_5	xy	y	2
ψ_6	x^3	x^2y	2
ψ_7	x^3y	y	2
ψ_8	x	x^2y	2

We omit the proof that each partial map ψ_i listed here actually extends to an automorphism. This would be an easy application of relatively free groups, which we chose to ignore in this article.

Let G again be arbitrary. Each a in G yields the *inner* automorphism t_a of G defined by $t_a(x) = axa^{-1}$. Elements v and w of G are *automorphic* if there is an automorphism of G mapping v to w. They are *conjugate* if there is an inner automorphism of G mapping v to w. Being conjugate is an equivalence relation in every group. For instance y and x^2y from K_8 are automorphic but not conjugate (all conjugacy classes of Abelian groups are singletons). We call a member v of G *characteristic* if it is fixed by all automorphisms. For example, in K_8 the element x^2 is characteristic as can be seen from Fact 4. For v in G its *centralizer* $C(v) := \{w \in G : wv = vw\}$ is a subgroup of G, and the *center* $Z(G) := \{w \in G : wv = vw \text{ for all } v \text{ in } G\}$ is a normal subgroup of G. If class(v) denotes the conjugacy class of v in G, then we record:

Fact 5. $|\,\mathrm{class}(v)| \cdot |C(v)| = |G|$.

Let p be a prime. If $|G| = p^n$ and v belongs to G, then Fact 5 implies that $|\,\mathrm{class}(v)| = p^i$ with $i \geq 0$. Because G is partitioned by its conjugacy classes and because $Z(G)$ is the union of all singleton classes, we infer:

Fact 6. If $|G| = p^n$ for a prime p, then p divides $|Z(G)|$.

We state for the record the classification of groups of order eight.

Theorem 1. *There are up to isomorphism exactly five groups of order eight. Besides $C_2 \times C_2 \times C_2$ they include the following:*

$$\begin{array}{lll} K_8 & (C_4, 2, \varphi_1, e), & (C_4, 2, \varphi_1, x^2) \\ D_8 & (C_4, 2, \varphi_2, e) & \\ C_8 & (C_4, 2, \varphi_1, x), & (C_4, 2, \varphi_1, x^3) \\ Q_8 & (C_4, 2, \varphi_2, x^2) & \end{array}$$

Every reader with a passing interest in groups has come across these chaps. In particular D_8 is the *dihedral* group of symmetries of the square, and $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ is the *quaternion* group with $i^2 = j^2 = k^2 = -1$, $i \cdot j = k$, $j \cdot i = -k$, etc. Written next to the groups for reference are their extension types, which will be defined in the next section.

Here is the paper's main theorem, whose proof will be divided into three lemmas.

Theorem 2. *There are up to isomorphism exactly fourteen groups of order sixteen. Besides the outsider $G_0 = C_2 \times C_2 \times C_2 \times C_2$ they can be listed as follows:*

$$\begin{array}{lll} G_1 = C_8 \times C_2 & (C_8, 2, \phi_1, e) & 16/2 \\ G_2 = SD_{16} = C_8 \overset{3}{\rtimes} C_2 & (C_8, 2, \phi_2, e) & 16/13 \\ & & \\ G_3 = C_8 \overset{5}{\rtimes} C_2 & (C_8, 2, \phi_3, e) & 16/11 \\ G_4 = D_{16} = C_8 \overset{7}{\rtimes} C_2 & (C_8, 2, \phi_4, e) & 16/12 \\ G_5 = Q_{16} & (C_8, 2, \phi_4, x^4) & 16/14 \\ G_6 = C_{16} & (C_8, 2, \phi_1, x) & 16/1 \\ & & \\ G_7 = K_4 \times C_4 & (K_8, 2, \psi_1, e) & 16/4 \\ G_8 = D_8 \times C_2 & (K_8, 2, \psi_3, e) & 16/6 \\ G_9 = K_4 \rtimes C_4 & (K_8, 2, \psi_5, e) & 16/9 \\ & & \\ G_{10} = Q_8 \overset{\tau}{\rtimes} C_2 & (K_8, 2, \psi_6, e) & 16/8 \\ G_{11} = Q_8 \times C_2 & (K_8, 2, \psi_3, x^2) & 16/7 \\ G_{12} = C_4 \rtimes C_4 & (K_8, 2, \psi_5, x^2) & 16/10 \\ G_{13} = C_4 \times C_4 & (K_8, 2, \psi_1, y) & 16/3 \end{array}$$

3. CYCLIC EXTENSIONS. Let $N \lhd G$ and $N' \lhd G'$. It is well known that from $N \simeq N'$ and $G/N \simeq G'/N'$ it does not follow that $G \simeq G'$. However, in the case of *cyclic* factor groups things are not too bad. Namely, suppose that $G/N \simeq C_n$. (Such a group G is called an (*inner*) *cyclic extension* of N.) Pick any a in G such that the coset Na has order n in G/N. Then $v = a^n$ is in N, and n is minimal with that property. Further, let τ in $\mathrm{Aut}(N)$ be the restriction of t_a to N. Then

$$\tau(v) = aa^n a^{-1} = a^n = v \tag{1}$$

and

$$\tau^n(x) = a \cdots a(axa^{-1})a^{-1} \cdots a^{-1} = a^n x a^{-n} = vxv^{-1} = t_v(x)$$

for all x in N, so

$$\tau^n = t_v. \tag{2}$$

In particular, when N is Abelian

$$\tau^n = \mathrm{id}\,. \tag{3}$$

The foregoing discussion motivates the following material.

Definition 1. A quadruple (N, n, τ, v) is an *extension type* if N is a group and if v in N and τ in $\mathrm{Aut}(N)$ are such that $\tau(v) = v$ and $\tau^n = t_v$.

Observe that this definition does not appeal to an element a anymore; in fact no enveloping group G of N is even mentioned. However, starting out with a group G having a factor group $G/N \simeq C_n$ it is clear from (1) and (2) that G *realizes* (obvious definition) at least one extension type (N, n, τ, v). Perhaps it realizes several (N, n, σ, w) by varying the *inducing* element a. The multiplication in G is pinned down by each fixed realized extension type (N, n, τ, v) as follows. Using the fact that each element g of G can be written in *normal form* as $g = xa^i$ with unique x in N and i in $\{0, 1, \ldots, n-1\}$, one has

$$(xa^i)(ya^j) = x(a^i y a^{-i})a^i a^j = x\tau^i(y)a^{i+j} = \begin{cases} (x\tau^i(y))a^{i+j} & \text{if} \quad i+j < n, \\ (x\tau^i(y)v)a^{i+j-n} & \text{if} \quad i+j \geq n. \end{cases} \tag{4}$$

Definition 2. The extension types (N, n, τ, v) and (N', n, σ, w) are *equivalent* if there is an isomorphism $\varphi : N \xrightarrow{\sim} N'$ such that $\sigma = \varphi \circ \tau \circ \varphi^{-1}$ and $w = \varphi(v)$.

One verifies that this is indeed an equivalence relation. For instance, the two extension types written to the right of C_8 in Theorem 1 are equivalent—choose $\varphi := \varphi_2$. Let (N, n, τ, v) be an extension type, and let $\varphi : N \xrightarrow{\sim} N'$ be any isomorphism. Using $(\varphi \circ \tau \circ \varphi^{-1})^n = \varphi \circ \tau^n \circ \varphi^{-1}$ one checks that $(N', n, \varphi \circ \tau \circ \varphi^{-1}, \varphi(v))$ is an extension type as well and is equivalent to (N, n, τ, v) by definition. In particular, if G realizes some extension type and if $G' \simeq G$, then G' realizes an equivalent extension type: just take φ to be the restriction to N of an isomorphism $\Phi : G \xrightarrow{\sim} G'$. The converse is more interesting and is stated in Lemma 1(a).

Lemma 1.

(a) *Groups G and G' that realize equivalent extension types must be isomorphic.*

(b) *Let v and w be automorphic elements of N, and let S be a fixed union of conjugacy classes of* $\mathrm{Aut}(N)$. *Suppose for τ ranging over S there is a group G_τ (respectively, F_τ) that realizes the extension type (N, n, τ, v) (respectively, (N, n, τ, w)). Then the families $\{G_\tau \mid \tau \in S\}$ and $\{F_\tau \mid \tau \in S\}$ comprise the same isomorphism types.*

(c) *Let v in N be characteristic, and let S be a conjugacy class of* $\mathrm{Aut}(N)$. *For any σ and τ in S any two groups G and G' realizing the extension types (N, n, σ, v) and (N, n, τ, v), respectively, are isomorphic.*

Proof.

(a) Let G and G' realize (N, n, τ, v) (respectively, (N', n, σ, w)), where $\sigma = \varphi \circ \tau \circ \varphi^{-1}$ and $w = \varphi(v)$ for some isomorphism $\varphi : N \xrightarrow{\sim} N'$. By definition there are elements a in $G \setminus N$ and b in $G' \setminus N'$ such that conjugation with them yields τ and σ, respectively, and such that $a^n = v$ and $b^n = w$. The map Φ from G to G' that assigns xa^i to $\varphi(x)b^i$ is well defined and bijective. In order to show that Φ is a homomorphism suppose that $i + j \geq n$ (the case $i + j < n$ is analogous). One computes

$$\begin{aligned}\Phi\big((xa^i)(ya^j)\big) &= \Phi\big(x(a^i y a^{-i})a^{i+j}\big) = \Phi\big(x\tau^i(y)va^{i+j-n}\big)\\ &= \varphi\big(x\tau^i(y)v\big)b^{i+j-n} = \varphi(x)\varphi\big(\tau^i(y)\big)\varphi(v)b^{i+j-n}.\end{aligned}$$

This is the same as

$$\begin{aligned}\Phi(xa^i)\Phi(ya^j) = \varphi(x)b^i\varphi(y)b^j &= \varphi(x)\big(b^i\varphi(y)b^{-i}\big)b^{i+j}\\ &= \varphi(x)\sigma^i\big(\varphi(y)\big)wb^{i+j-n},\end{aligned}$$

since $\sigma^i = \varphi \circ \tau^i \circ \varphi^{-1}$ implies that $\sigma^i \circ \varphi = \varphi \circ \tau^i$.

(b) Pick φ from $\mathrm{Aut}(N)$ with $\varphi(v) = w$. Obviously the correspondence given by $\tau \mapsto \sigma = \varphi \circ \tau \circ \varphi^{-1}$ is a bijection of S. Because the extension types (N, n, τ, v) and (N, n, σ, w) are equivalent, corresponding realizing groups G_τ and F_σ must be isomorphic by (a).

(c) Let τ in S be fixed. As ϕ runs through $\mathrm{Aut}(N)$, $\phi \circ \tau \circ \phi^{-1}$ runs through S. By (a) all groups of extension type $(N, n, \phi \circ \tau \circ \phi^{-1}, \phi(v))$ are isomorphic. But $\phi(v) = v$ for all ϕ since v is characteristic. ■

Let G be a group of order eight. If all elements of G have order at most two, then $G \simeq C_2 \times C_2 \times C_2 = (C_2)^3$ by Fact 2. If $G \not\simeq (C_2)^3$, there is some x in G with $\langle x \rangle = C_4$ (Why?). Since subgroups of index two are always normal one has $C_4 \lhd G$. Since $|C_4| \cdot |\mathrm{Aut}(C_4)| = 8$ by Fact 3, there can be at most eight extension types $(C_4, 2, \varphi_i, x^j)$; in fact there are exactly six because x and x^3 are not fixed by φ_2. These extension types are listed in Theorem 1, along with four groups that happen to realize them. For instance, setting $C_4 = \langle x \rangle = \langle i \rangle$ one verifies that all elements a in $Q_8 \setminus C_4$ induce the same extension type $(C_4, 2, \varphi_2, x^2)$. On the other hand, by Lemma 1(a), *every* group G of order eight with $C_4 \lhd G$ must be isomorphic to one of the four groups in Theorem 1. Hence, provided K_8, D_8, C_8, and Q_8 are mutually nonisomorphic, there are *exactly* four groups G of order eight that are not isomorphic to $(C_2)^3$. Looking at the number of elements of order two, four, and eight in each group shows that K_8, D_8, C_8, and Q_8 are indeed mutually nonisomorphic. This establishes Theorem 1.

The proof of Theorem 2 proceeds along the same lines but with two obstacles. First, a group $G \not\simeq (C_2)^4$ of order sixteen need not have a subgroup C_8. But if it hasn't, then K_8 turns out to be a decent substitute (Lemma 2). Second, in the context of Theorem 2 it is unclear whether the extension types derived in Lemma 3, aided by Lemma 1, can actually be realized. That problem will be dealt with in section 4. The isomorphism problem is handled in section 5.

Lemma 2. *If G is a group of order sixteen that is not isomorphic to $C_2 \times C_2 \times C_2 \times C_2 = (C_2)^4$, then either $C_8 \lhd G$ or $K_8 \lhd G$.*

Proof. We assume that G has no element of order eight and show that K_8 occurs as a subgroup of G. By assumption $G \not\simeq (C_2)^4$, so Fact 2 guarantees that there are elements of order four in G. By Fact 6 we can fix an element z of order two in $Z(G)$ and write $H = \langle z \rangle$. Clearly $H \lhd G$. We consider two cases.

Case 1: There is an element x in G of order four with $x^2 \neq z$. Then $\langle x \rangle \cap H = \{e\}$, so $\langle x, z \rangle$ is K_8 by Fact 1.

Case 2: All elements x in G of order four have $x^2 = z$. Then all elements of G/H have order at most two, so G/H is Abelian by Fact 2. Now let x have order four. Since G/H is Abelian, all conjugates of x lie in the coset Hx. Thus $|\operatorname{class}(x)| \leq 2$, so in light of Fact 5 the centralizer $C = C(x)$ has order at least eight. It follows that there exists y in C with $y \notin \langle x \rangle$. If $o(y) = 2$, then $\langle x, y \rangle = K_8$. If $o(y) = 4$, then $y^2 = z$, whence $(xy)^2 = x^2y^2 = zz = e$ and $o(xy) \leq 2$. If $o(xy) = 1$, then $y = x^{-1} \in \langle x \rangle$, which is not the case. Thus xy is an element of order two that commutes with x. Therefore, provided xy is not a member of $\langle x \rangle$, we are done. Fortunately the assumption that xy belongs to $\langle x \rangle$ implies that $xy = x^2$, which delivers the contradiction $y = x$. ■

Lemma 3. *Each group G of order sixteen that is not isomorphic to $(C_2)^4$ realizes one of the extension types listed in Theorem 2 (but possibly not every extension type in Theorem 2 is realized).*

Proof. In the proof of Theorem 1 we saw that there are six extension types $(C_4, 2, \tau, v)$. Similarly one argues, using Fact 3 (respectively, Fact 4) that there are sixteen extension types $(C_8, 2, \tau, v)$ and thirty-two extension types $(K_8, 2, \tau, v)$. By Lemma 2 each group $G \not\simeq (C_2)^4$ of order 16 realizes (up to equivalence) at least one of these $16 + 32$ extension types. Observe that some of the forty-eight extension types may be equivalent to each other, and some a priori not realizable at all. Lemma 3 asserts that, in any case, each G realizes one of the merely thirteen extension types listed in Theorem 2.

Putting $C_8 = \langle x \rangle$ we first derive the six extension types $(C_8, 2, \tau, v)$ by looking at the minimal order of an inducing element a in $G \setminus C_8$. In the sequel $v = a^2$ and τ is the restriction to C_8 of the conjugation map t_a. The breakdown into six cases goes as follows.

Case 1: $o(a) = 2$. Then $v = e$. By Fact 3 all ϕ_i in $\operatorname{Aut}(C_8)$ satisfy (1) and (3). Thus we get the extension types $(C_8, 2, \phi_i, e)$ $(1 \leq i \leq 4)$. Henceforth we may assume that $o(b) \geq 4$ for all b in $G \setminus C_8$, since otherwise we are back to case 1.

Case 2: $o(a) = 4$. Here $v = x^4$. If $\tau = \phi_2$, then $(xa)(xa) = x\phi_2(x)a^2 = xx^3a^2 = e$ by (4), so $o(xa) = 2$ contrary to our assumption. Similarly, if τ is ϕ_1 or ϕ_3, then $o(x^2a) = 2$. Thus at most $(C_8, 2, \phi_4, x^4)$ is a new extension type.

Case 3: $o(a) = 8$. In this situation $v = x^2$ or $v = x^6$. Either way $\tau(v) = v$ occurs only for τ in $S = \{\phi_1, \phi_3\}$. Let $v = x^2$. Then $o(x^3a) = 2$ for $\tau = \phi_1$, and $o(xa) = 2$ for $\tau = \phi_3$. If $v = x^6$, then by Lemma 1(b) the same groups arise since x^6 is automorphic to x^2.

Case 4: $o(a) = 16$. Then $G \simeq C_{16}$ which, for instance, realizes the extension type $(C_8, 2, \phi_1, x)$.

We now derive the seven extension types $(K_8, 2, \tau, v)$, where

$$K_8 = \{e, x, x^2, x^3, y, xy, x^2y, x^3y\}$$

as usual. By the previous discussion we may assume that $o(z) < 8$ holds for all z in G. By Fact 4 we have $\mathrm{Aut}(K_8) = \{\psi_1, \cdots, \psi_8\}$ and the τ with $\tau^2 = \mathrm{id}$ are the elements of the conjugacy classes $\{\psi_1\}$, $\{\psi_3\}$, $\{\psi_5, \psi_7\}$, and $\{\psi_6, \psi_8\}$. We proceed again according to the minimal order of an inducing element a in $G \setminus K_8$ and put $v = a^2$. Here τ is the restriction of t_a to K_8.

Case 5: $o(a) = 2$. Then $v = e$ is characteristic, so by Lemma 1(c) it suffices to consider any representatives of the conjugacy classes just indicated. Thus we get the extension types $(K_8, 2, \psi_i, e)$ with (say) $i = 1, 3, 5$, or 6.

Case 6: $o(a) = 4$. In this instance v is one of x^2, y, or x^2y. In view of case 5 we may assume that $o(b) \geq 4$ for all b in $G \setminus K_8$. We consider three subcases.

i. $v = x^2$. Since v is characteristic it suffices again to focus on ψ_1, ψ_3, ψ_5, and ψ_6. If $\tau = \psi_1$, then $o(xa) = 2$, while if $\tau = \psi_6$, then $o(xya) = 2$. Thus we are left with $(K_8, 2, \psi_3, x^2)$ and $(K_8, 2, \psi_5, x^2)$.

ii. $v = y$. Because we must have $\tau(v) = v$ only $\tau = \psi_1, \psi_3, \psi_5$, or ψ_7 qualifies. If $\tau = \psi_7$, then $o(xa) = 2$. If $\tau = \psi_5$, then

$$(xa)(xa) = x\psi_5(x)a^2 = x(xy)y = x^2,$$

so putting $\overline{a} = xa$ we are back in subcase (i). If $\tau = \psi_3$, then $ax^2a^{-1} = \psi_3(x^2) = x^2$, so a and x^2 commute, whence

$$N := \langle a, x^2 \rangle = \{e, a, y, ya, x^2, x^2a, x^2y, x^2ya\}$$

is isomorphic to $K_8 = \langle x, y \rangle$ with y being the unique characteristic nonidentity element of N (Fact 1). Since

$$(xa)(xa) = x\psi_3(x)a^2 = x(x^3)y = y,$$

setting $\overline{a} = xa \notin N$ brings us back to subcase (i) again. Therefore we are left with $(K_8, 2, \psi_1, y)$.

iii. $v = x^2y$. Because we must have $\tau(v) = v$, only the mappings τ in $S = \{\psi_1, \psi_3, \psi_5, \psi_7\}$ qualify. Because v is automorphic to y by Fact 4, and because S is a union of conjugacy classes, Lemma 1(b) predicts that we end up with the same groups as in subcase (ii). ■

4. THE BREATH OF LIFE. Here we give life to the extension types listed in Theorem 2. Thus we show that each extension type is actually realized by some group. We do so in two ways: first, by constructing permutation groups that do the job (the pedestrian approach); second, by appealing to the cyclic extension theorem (the proof of which, however, is omitted).

As to the first approach, say there *is* a group G_2 of extension type $(C_8, 2, \phi_2, e)$. Then its elements are $e, x, x^2, \cdots, x^7, a, xa, \cdots, x^7a$, and according to (4) the multiplication is given by

$$(x^k a^i)(x^\ell a^j) = \begin{cases} x^k x^{3\ell} a & \text{if } i + j = 1, \\ x^k x^{3\ell} & \text{otherwise.} \end{cases} \tag{5}$$

We emphasize that we cannot conversely simply take a sixteen-element set, label its elements x^i and $x^i a$ $(0 \leq i \leq 7)$, and *define* an operation by (5) because who assures us of associativity?[1] Rather, recall that the Cayley representation of an arbitrary group G

[1]That person is Otto Hölder, but be patient: we are still pedestrians.

of order m assigns to g in G the permutation ρ_g in S_m defined by $\rho_g(z) = zg$. Here S_m is the symmetric group on the set G. If one composes permutations from left to right, then $\rho_g \circ \rho_h = \rho_{gh}$. Applying the Cayley representation to the generators x and a of our hypothetical group G_2 and using cycle notation for permutations yields

$$\rho_x = (e, x, x^2, x^3, x^4, x^5, x^6, x^7)\ (a, x^3a, x^6a, xa, x^4a, x^7a, x^2a, x^5a),$$
$$\rho_a = (e, a)\ (x, xa)\ (x^2, x^2a)\ (x^3, x^3a)\ (x^4, x^4a)\ (x^5, x^5a)\ (x^6, x^6a)\ (x^7, x^7a).$$

Relabelling puts these into more recognizable form, namely,

$$\xi = (1, 2, 3, 4, 5, 6, 7, 8)\ (9, 10, 11, 12, 13, 14, 15, 16),$$
$$\alpha = (1, 9)\ (2, 12)\ (3, 15)\ (4, 10)\ (5, 13)\ (6, 16)\ (7, 11)\ (8, 14)$$

Forgetting about the origin of the latter permutations, consider the subgroup $H = \langle \xi, \alpha \rangle$ of S_{16}. By calculation one verifies that $\alpha\xi\alpha^{-1} = \xi^3$, whence $\alpha\xi^k\alpha^{-1} = (\alpha\xi\alpha^{-1})^k = \xi^{3k}$, so $\alpha\xi^k = \xi^{3k}\alpha$. Together with $\xi^8 = \alpha^2 = \text{id}$ this implies that each element of H can be written as $\xi^k\alpha^i$ with $0 \le k \le 7$ and $0 \le i \le 1$. For instance,

$$\xi^4\alpha\xi^7\alpha\xi^2\alpha = \xi^4\alpha\xi^7\xi^6\alpha\alpha = \xi^4\alpha\xi^5 = \xi^4\xi^{15}\alpha = \xi^3\alpha.$$

We conclude that $|H| \le 16$. On the other hand, again by brute-force calculation, one sees that the sixteen pairs (k, i) yield different elements $\xi^k\alpha^i$. Hence $|H| = 16$, and it follows from $\alpha^2 = e$ and $\alpha\xi\alpha^{-1} = \xi^3$ that α in $H \setminus \langle\xi\rangle$ induces the extension type $(C_8, 2, \phi_2, e)$.

Remarkably, *each* extension type (N, n, τ, v) can be realized by a group of permutations crafted in analogous fashion (this provides nice exercises for students). That this method always succeeds is a consequence of Cayley's theorem in tandem with the following result:

Theorem 3 (Cyclic Extension Theorem). *Each extension type (N, n, τ, v) is realized by a suitable group G.*

The elements of G are the ordered pairs (x, a^i) with x in N and $0 \le i < n$. Here a is just a symbol. Define a binary operation as follows:

$$(x, a^i) * (y, a^j) = \begin{cases} (x\tau^i(y), a^{i+j}) & \text{if} \quad i + j < n, \\ (x\tau^i(y)v, a^{i+j-n}) & \text{if} \quad i + j \ge n. \end{cases} \tag{6}$$

This definition is reminiscent of (4), yet it now becomes a nontrivial matter to establish the associativity of $*$ (see the remarks in section 6)! This group G of ordered pairs is sometimes called an *outer* cyclic extension of N, as opposed to the inner cyclic extensions discussed in section 3. Once G is established as group one usually writes a for (e, a), v for (v, e), and generally xa^i for (x, a^i). Note that τ now coincides with the restriction of $x \mapsto axa^{-1}$ to N.

Let us look at some special cases. Setting $C_n = \langle a \rangle$ one has $v = e$ if and only if $N \cap C_n = \{e\}$. In this case (6) defines a so-called semidirect product $N \overset{\tau}{\rtimes} C_n$. If additionally $\tau = \text{id}$, then (6) boils down to the familiar direct product $N \times C_n$. We write $G = N \rtimes C_n$ if (up to isomorphism) G is the *only proper* semidirect product of N and C_n, meaning the only one besides $N \times C_n$. If $G = N \overset{\tau}{\rtimes} C_n$ with $N = C_m$, then we write $G = C_m \overset{k}{\rtimes} C_n$. Here k is defined by the fact that τ is the map $z \mapsto z^k$ of C_m to itself.

All our G_i $(1 \leq i \leq 13)$ were constructed as cyclic extensions of C_8 or K_8 but, once found, most of them admit simpler descriptions. For instance, G_{13} contains the subgroups $H_1 = \langle x \rangle$ and $H_2 = \langle a \rangle = \{a, y, ya, e\}$, where $xa = ax$. Thus $G_{13} \simeq H_1 \times H_2 = C_4 \times C_4$ by Fact 1. Similarly $xy = yx$ and $ay = ya$ in G_{11}, so $G_{11} \simeq \langle x, a \rangle \times \langle y \rangle$. Because $\langle x, a \rangle$, via the inducing element a, realizes the extension type $(\langle x \rangle, 2, \varphi_2, x^2)$ that is listed next to Q_8 in Theorem 1, one concludes that $G_{11} \simeq Q_8 \times C_2$. Similarly G_{10} contains the subgroup $\langle x, ya \rangle \simeq Q_8$ that is disjoint from $\langle y \rangle \simeq C_2$. Thus $G_{10} \simeq Q_8 \overset{\tau}{\rtimes} C_2$, which turns out to be the only proper semidirect product of Q_8 and C_2 without elements of order eight. Further

$$G_9 \simeq \langle y, a \rangle \overset{\tau}{\rtimes} \langle x \rangle = K_4 \overset{\tau}{\rtimes} C_4,$$

where $\tau(y) = y$ and $\tau(a) = ya$. Because all order two elements σ of $\mathrm{Aut}(K_4) \simeq S_3$ are conjugate, they yield isomorphic semidirect products $K_4 \overset{\sigma}{\rtimes} C_4$ by Lemma 1(c). Hence the notation $G_9 = K_4 \rtimes C_4$. We leave it to the reader to verify the other entries in the second column of Theorem 2. Notice that $D_{16} = C_8 \overset{7}{\rtimes} C_2$ is a *dihedral*, $SD_{16} = C_8 \overset{3}{\rtimes} C_2$ a *semi-dihedral*, and Q_{16} is a *generalized quaternion* group, the second of an infinite family Q_{2^n} $(n \geq 3)$.

5. THE ISOMORPHISM PROBLEM. By Lemma 1(a) and Theorem 3 equivalent extension types are *always* realized by isomorphic groups. Unfortunately inequivalent extension types (N, n, τ, v) and (N', n, σ, w) may *sometimes* be realized by isomorphic groups G and G' as well. (This happens if and only if each isomorphism $\Phi : G \xrightarrow{\sim} G'$ either has $\Phi(v) \neq w$ or $(\Phi|N) \circ \tau \circ (\Phi|N)^{-1} \neq \sigma$; the group $G = G' = K_8$ in Theorem 1 is a case in point.) This cumbersome situation is referred to as the *isomorphism problem*. One way to prove that two groups are not isomorphic is to show that one has more elements of a certain order than the other. Thus, we consider first the orders of all elements of our groups:

Table 1.

	x	x^2	x^3	x^4	x^5	x^6	x^7	a	xa	x^2a	x^3a	x^4a	x^5a	x^6a	x^7a
G_1	8	4	8	2	8	4	8	2	8	4	8	2	8	4	8
G_2	8	4	8	2	8	4	8	2	4	2	4	2	4	2	4
G_3	8	4	8	2	8	4	8	2	8	4	8	2	8	4	8
G_4	8	4	8	2	8	4	8	2	2	2	2	2	2	2	2
G_5	8	4	8	2	8	4	8	4	4	4	4	4	4	4	4
G_6	8	4	8	2	8	4	8	16	16	16	16	16	16	16	16
	x	x^2	x^3	y	xy	x^2y	x^3y	a	xa	x^2a	x^3a	ya	xya	x^2ya	x^3ya
G_7	4	2	4	2	4	2	4	2	4	2	4	2	4	2	4
G_8	4	2	4	2	4	2	4	2	2	2	2	2	2	2	2
G_9	4	2	4	2	4	2	4	2	4	2	4	2	4	2	4
G_{10}	4	2	4	2	4	2	4	2	2	2	2	4	4	4	4
G_{11}	4	2	4	2	4	2	4	4	4	4	4	4	4	4	4
G_{12}	4	2	4	2	4	2	4	4	4	4	4	4	4	4	4
G_{13}	4	2	4	2	4	2	4	4	4	4	4	4	4	4	4

All nonidentity elements of $G_0 = (C_2)^4$ have order 2, so G_0 is certainly not isomorphic to any other G_i. Counting the number of elements of order two, four, eight, and sixteen, respectively, it follows that among $G_1, \ldots, G_6$ at most $G_1 \simeq G_3$ could occur. But that's impossible since one is Abelian and the other is not. Because of the number of elements of order eight, no G_i $(1 \leq i \leq 6)$ is isomorphic to any G_j $(7 \leq j \leq 13)$. As to the second set of groups, it remains to show that no two of G_7, G_9, and G_{10} are isomorphic and that no two of G_{11}, G_{12}, and G_{13} are isomorphic. Now G_7 is Abelian but the non-Abelain G_9 and G_{10} are more subtle to distinguish: in G_9 the two subgroups $\{xa, x^2y, x^3ya, e\}$ and $\{x, x^2, x^3, e\}$ intersect trivially, whereas in G_{10} every four-element subgroup contains x^2 (exercise). Similarly G_{13} is Abelian but again the non-Abelian G_{11} and G_{12} are harder to tell apart. Taking Q_8 as $\{\pm 1, \pm i, \pm j, \pm k\}$ the center of $G_{11} \simeq Q_8 \times C_2$ is $\{(1, e), (1, a), (-1, e), (-1, a)\}$ and contains only one nonunit square $(-1, e) = (i, e)^2$. On the other hand, $Z(G_{12})$ contains two nonidentity squares x^2 and $y = (xa)^2$.

Despite our focus on C_8 and $K_8 = C_4 \times C_2$, a given group of order sixteen can of course be a cyclic extension of $(C_2)^3$ or D_8 or Q_8 as well. Without proof, Table 2 gives the cyclic extensions of the five groups of order eight:

Table 2.

	G_0	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8	G_9	G_{10}	G_{11}	G_{12}	G_{13}
C_8		x	x	x	x	x	x							
K_8		x		x				x	x	x	x	x	x	x
$C_2 \times C_2 \times C_2$	x							x	x	x				
D_8			x		x				x		x			
Q_8			x			x					x	x		

6. HISTORICAL REMARKS. Hölder initiated the theory of group extensions in his seminal article "Bildung zusammengesetzter Gruppen" [**5**], which featured the cyclic extension theorem (i.e., Theorem 3, but somewhat disguised). The groups that can be obtained from the trivial group by iteratively building cyclic extensions are called *solvable*. Statistically speaking "most" groups are solvable. For instance, all groups of order less than sixty are solvable. This pleasant state of affairs notwithstanding, Schreier, Artin, and others deemed it necessary to "extend" Hölder's extension theory to noncyclic factor groups G/N. The prize is a barrage of technicalities that rather obscures the essentials. Deplorably most texts on group theory that deal with extension theory invert the arrow of history and mention Theorem 3 at most as a corollary of Artin-Schreier theory—that at a time when the novice has long since given up.

After this praise of the cyclic extension theorem, we should say a few words about *p-groups* (i.e., groups whose order p^e is a power of a prime p). According to Sylow's main theorem each finite group G of cardinality $n = p_1^{e_1} \cdots p_k^{e_k}$ contains a subgroup H_i of order $p_i^{e_i}$ for $i = 1, 2, \ldots, k$, and the structure of the H_i determines the structure of G to a large extent. Therefore it is important to classify p-groups. The groups of order p^3 and p^4 were classified by Hölder [**4**] along with the groups of order p^2q and pqr. The groups of order 2^n $(n \leq 6)$ are given in [**6**]. More recently, the classification of small groups has advanced tremendously through the work of Besche, Eick, and O'Brien. Now at last humankind knows that there are 49,487,365,422 groups of order 2^{10}, about seven for each human being. We recommend the survey article [**2**], where one finds extensive historical background and a description of the construction and enumeration algorithms. A table of the number of groups of order at most two thousand, and much more, can be found in [**3**]. (Exercise: How many of these numbers were not known to Hölder?) One tenth of this table is reproduced here as Table 3.

Table 3.

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9
0		1	1	1	2	1	2	1	5	2
10	2	1	5	1	2	1	14	1	5	1
20	5	2	2	1	15	2	2	5	4	1
30	4	1	51	1	2	1	14	1	2	2
40	14	1	6	1	4	2	2	1	52	2
50	5	1	5	1	15	2	13	2	2	1
60	13	1	2	4	267	1	4	1	5	1
70	4	1	50	1	2	3	4	1	6	1
80	52	15	2	1	15	1	2	1	12	1
90	10	1	4	2	2	1	231	1	5	2
100	16	1	4	1	14	2	2	1	45	1
110	6	2	43	1	6	1	5	4	2	1
120	47	2	2	1	4	5	16	1	2328	2
130	4	1	10	1	2	5	15	1	4	1
140	11	1	2	1	197	1	2	6	5	1
150	13	1	12	2	4	2	18	1	2	1
160	238	1	55	1	5	2	2	1	57	2
170	4	5	4	1	4	2	42	1	2	1
180	37	1	4	2	12	1	6	1	4	13
190	4	1	1543	1	2	2	12	1	10	1

We also mention [**9**], which contains the multiplication tables of all groups of order thirty-two or less, along with further information such as the structure of the automorphism group, the lattice of subgroups, and the character table. No proofs are given in [**9**]. The last column in Theorem 2 displays the notation used in [**9**].

ACKNOWLEDGMENT. I am grateful to the referee for enhancing the proof of Lemma 2 and for advancing the cyclic extension theorem rather than generators and relations.

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Algebra 1

When I first started algebra,
I never ever thought
that I had signed up
for a baby-sitting job.

Working with those signs
is like watching little kids.
I wish that I could put the signs
in jars and screw the lids.

The problem with those signs is,
like kids, they like to play.
Soon as I track down little x,
subtraction runs away.

And meanwhile, on the other side
of my graph paper,
cross-out lines have tied up
a baby integer.

Oh how I try to save her,
but was she 1 or 10?
Try as I might,
she won't be back again.

I really should start over,
but I've done a lot of work.
How painful it will be!
But I will not shirk.

I'll start fresh. I will do it,
though it will take long.
At least I will be certain that
the problem won't be wrong.

The good thing about algebra:
You get a second chance.
With kids, if you mess up,
there is no penitence.

——Submitted by Ben Yandell, Cal Tech; composed by his (then) fourteen-year-old daughter Kate. Tragically, Ben passed away of a heart attack on August 25, 2004.